PAPER_1613_NUCLEAR_U238_BE_A_7_570

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PAPER_1613 — U-238 Binding Energy per A = U-238 BE/A = $F\cdot K■ + \beta^2 + \beta^3 + F\cdot\beta + 3 \approx 7.568$ MeV

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Nuclear
Date: June 18, 2026
Status: CLOSED — 0.033% residual

Observation

PAPER_1209II_S667 (Nuclear Unified Proof Set) gives the closed-form expression:

```
U-238 BE/A = F·K■ + β² + β³ + F·β + 3 ≈ 7.568 MeV
```

Residual to AME2020: **0.033%**.

UQFF Closed Identity

```
U-238 BE/A = F·K■ + β² + β³ + F·β + 3 ≈ 7.568 MeV 0.033%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental nuclear measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209II_S667
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "nuclear_u238_be_a_7_570"})`
